

RFI — Feed & water intake sensing for equine monitoring (BSV EquiCare)

To: Feed / water metering vendor or systems integrator

From: Bharat Sports Venture — EquiCare engineering

Covers client monitoring points: 9 (watering — quantity, pattern, time), 10 (feeding — quantity, pattern, time)

0. How to reply — please read first

Please answer all of this in ONE consolidated reply. We are evaluating several suppliers in parallel on a fixed timeline and will not be sending rounds of follow-up questions. In practice that means:

1. **Answer every numbered question**, in the reply slots provided. If something is not supported, **write that plainly** — "not supported" is a perfectly good answer and costs you nothing. A blank or skipped item, however, we have to record as *not supported*, because we cannot tell the difference.
2. **Attach the documents and files listed in the checklist at the end.** Please send them with this reply, not "on request" — a claim we cannot verify counts as unverified.
3. **Send the actual artifact, not a description of it.** Our single worst experience with a previous supplier: they confirmed "Linux SDK — yes", and the package that arrived was x86-64 only, with no ARM64 build available at all. That cost both sides weeks. So where we ask for a binary, driver, register map or protocol spec, please attach the real file (or a download link) — or state clearly that it does not exist.
4. If a question does not apply to your product, write **"N/A"** and one line saying why.

We would rather receive an honest reply with several "not supported" answers than an optimistic one that unravels at integration.

1. What we are building

A 24/7 monitoring system for stabled horses in India. Each stall has sensors feeding an **on-site edge computer** (NVIDIA Jetson). A change in water or feed intake is one of the earliest indicators of illness, so we need **per-horse, timestamped consumption** — not daily totals.

All analytics and alerting are ours. We need reliable measurement plus a documented way to read it locally. We are open to either **inline flow metering** or **load-cell (weighed) vessels** — and to buying **standard industrial components** rather than a packaged product, if that gives us cleaner data access.

2. Hard requirements — please confirm these first

Pass/fail for us. If any cannot be met, please say so plainly rather than leaving it blank.

#	Requirement	Why
H1	Open, documented local protocol — Modbus RTU/TCP, RS-485, pulse/dry-contact, 4–20 mA or MQTT — with the register map / protocol document supplied with your reply .	We integrate directly on our edge box.
H2	Linux ARM64 (aarch64 / NVIDIA Jetson) . If you supply a driver or library it must be built and tested for aarch64 — not Windows/.NET-only, not x86-64-only . Open protocols make this a non-issue; tell us which applies.	A previous vendor shipped x86-only binaries and could not rebuild — weeks lost.
H3	No mandatory vendor cloud . All data readable on the local network.	India DPDP-2023 residency; unreliable rural internet.
H4	Per-horse attribution — a dedicated per-stall vessel, or RFID identification at a shared vessel.	Per-horse trends are the entire clinical value. A shared trough with no animal ID is unusable for us.
H5	Wash-down rated with safe wetted materials — IP66/IP67 on housing <i>and</i> connectors/glands, corrosion-resistant, food/potable-grade parts in contact with water or feed.	Daily high-pressure wash-down; animal safety.

REPLY (H1–H5)

3. WATER — measurement

Answer with **specific values**, not "yes — supported". Please answer for whichever approach you supply, and say which you would recommend for horses.

- Measurement principle** — electromagnetic / ultrasonic / hall-effect / turbine inline meter, or load-cell vessel? *We prefer no or low moving parts.*

REPLY

- Resolution and accuracy**. Flow meter: **ml per pulse** and accuracy across **0.1–8 L/min**. Load cell: **resolution in grams**, capacity, and drift over 24 h.

REPLY

3. **Minimum measurable flow / cut-off**, in L/min and ml/min. A horse taking slow sips must not read as zero — state the threshold below which nothing is reported.

REPLY

4. **Indian water conditions** — tolerance to **high-TDS groundwater and scaling**, plus hay debris and bio-fouling. State the cleaning/service interval.

REPLY

5. **Refill vs. consumption**. Can the device distinguish **water being drunk** from **the vessel being refilled**, and log them as **separate timestamped events**? This is essential — otherwise a refill looks like a horse drinking 20 L in one go.

REPLY

6. **Load-cell option**: can it separate **drinking** from **spillage, evaporation and play**? If not, state the expected daily error from evaporation.

REPLY

7. **Load-cell sampling rate and debounce** — horses nudge, lean on and bite buckets. How are those transient spikes rejected without losing genuine drinking events?

REPLY

4. FEED — measurement

8. **Dose resolution and accuracy in grams**, stated separately for **pellets**, **textured / sweet feed** and **loose grain** if they differ. Does dosing hold accuracy in **monsoon humidity**?

REPLY

9. **Weigh-back / refusal**. Can the system measure **what the horse actually ate** (offered minus leftover), or only what was dispensed? *Refusal is our single strongest early-illness signal, so a dispense-only feeder is of limited use to us.* State how leftovers are measured.

REPLY

10. **Forage / hay**. Do you offer a **load-cell manger or haynet**? State resolution and capacity, and whether consumption can be separated from wastage. *Horses eat mostly forage, so a concentrate-only feeder covers the minority of intake.*

REPLY

11. **Programmable schedule per horse** (morning / midday / evening / free-choice), updatable **via the local API** — not only a front panel or cloud portal.

REPLY

12. **Dispense confirmation** — does each dispense emit a record with scheduled vs. actual time, grams delivered and status?

REPLY

13. **Fault reporting** — are **hopper empty**, **bridging/jam**, **motor stall** and **under/over-run** reported as **distinct alarms** over the protocol? Please list the fault codes.

REPLY

5. Attribution

14. **Dedicated per-stall vessel, or shared?** If shared, how is intake attributed to an individual animal?

REPLY

15. **If RFID:** which standard — **LF 134.2 kHz ISO equine microchip**, UHF, or BLE? State read reliability *while the animal is actually feeding* (head down, moving), and confirm the animal ID is emitted with **every** event.

REPLY

6. Integration

16. **Interfaces available** — Modbus RTU/RS-485, Modbus TCP, pulse/dry-contact, 4–20 mA, BLE, Wi-Fi, Ethernet, MQTT. **Please attach the full register map**, baud rates and devices-per-bus limit.

REPLY

17. **Event model.** Do you output **continuous totals**, or **discrete bouts** with start, end, duration and volume/grams? *We need timestamped bouts to analyse pattern and timing, not just a daily counter.*

REPLY

18. **Timestamping** — onboard RTC or edge-stamped? State RTC sync method (**PTP preferred, NTP acceptable**) and drift over 24 h. *We fuse this with video, thermal and audio.*

REPLY

19. **Local buffering and backfill** — ≥ 24 h onboard, replayed with original timestamps after a network or power outage. State onboard depth.

REPLY

20. **Per-reading status/confidence flag** — in-range, settled vs. in-motion, dispense verified, sensor fault?

REPLY

21. **Devices per edge box** — RS-485 addressing limits and throughput when ingesting water + feed from multiple stalls on one bus.

REPLY

7. Calibration, environment and operations

22. **Calibration** procedure and frequency; can barn staff do it in the field; is it accessible from Linux over the protocol?

REPLY

23. **Documented drift spec** (zero and span, over temperature and time), **auto-tare between fills**, and temperature effect across **5–50 °C** with any onboard compensation.

REPLY

24. **Mechanical abuse rating** — kick, lean, crib, rub. Is a **guard or recessed mount** available, with electronics out of the kick/chew/wash zone?

REPLY

25. **Hygiene** — are water/feed-contact parts **removable and disinfectant-safe daily without re-calibration?**

REPLY

26. **Power** — wired DC, PoE (state class and watts), mains or battery? *If PoE, note it shares the barn switch budget with our cameras.*

REPLY

27. **Vessel capacity** (litres of water / kg of feed), and can a **manual refill be logged as an event** with operator identity and time?

REPLY

28. **Remote firmware/config update** from the edge (schedules, recalibration) without per-stall physical access; does it work offline?

REPLY

8. Interim manual logging

29. Sensors will not be installed on day one. Can manually-entered feed/water amounts be recorded through the same interface, carrying the same schema (`{..., source: 'manual', confidence}`), so the eventual sensor upgrade is seamless? *If not, we will handle this in our own software — just confirm.*

REPLY

9. Evaluation

We would like **1 water sensor + 1 weigh-back feeder** as evaluation units, with the **protocol documentation / register map and sample code**, so we can validate on our Jetson bench before committing to a design.

REPLY (AVAILABILITY + LEAD TIME FOR EVALUATION HARDWARE)

10. Optional — field validation

Any **field-validation data with horses or comparable livestock** (real-world accuracy sustained over weeks, not a lab figure) would be valuable to us.

REPLY

Response checklist

Please confirm each item is attached or answered, so we can evaluate in a single pass:

	Item
☐	Every numbered question answered (or marked "not supported" / "N/A")
☐	All hard requirements (H-numbers) explicitly confirmed or declined
☐	Sensor / component datasheets

	Item
☐	Protocol document, register map or API reference — the actual file
☐	ARM64 (aarch64) build attached or download link — <i>or</i> a plain statement that none exists
☐	Sample data payload showing real field names and units
☐	Sample code, if you provide any
☐	Evaluation-unit availability and lead time
☐	A named technical contact we can reach directly with any single clarification

If anything on this list genuinely cannot be shared before an order, please say which item and why, so we can factor it in rather than assume the worst.

Commercial terms are not part of this RFI — we will discuss those separately once the technical fit is confirmed.